

Focused ultrasound and microbubble-mediated delivery of CRISPR-Cas9 ribonucleoprotein to human induced pluripotent stem cells

Kyle Hazel,¹ Davindra Singh,¹ Stephanie He,¹ Zakary Guertin,¹ Mathieu C. Husser,¹ and Brandon Helfield^{1,2}

¹Department of Biology, Concordia University, 7141 Sherbrooke St. W, H4B 1R6 Montreal, QC, Canada; ²Department of Physics, Concordia University, 7141 Sherbrooke St. W, H4B 1R6 Montreal, QC, Canada

CRISPR-Cas9 ribonucleoproteins (RNPs) have been heavily considered for gene therapy due to their high on-target efficiency, rapid activity, and lack of insertional mutagenesis relative to other CRISPR-Cas9 delivery formats. Genetic diseases such as hypertrophic cardiomyopathy currently lack effective treatment strategies and are prime targets for CRISPR-Cas9 gene editing technology. However, current *in vivo* delivery strategies for Cas9 pose risks of unwanted immunogenic responses. This proof-of-concept study aimed to demonstrate that focused ultrasound (FUS) in combination with microbubbles can be used to deliver Cas9-sgRNA (single-guide RNA) RNPs and functionally edit human induced pluripotent stem cells (hiPSCs) *in vitro*, a model system that can be expanded to cardiovascular research via hiPSC-derived cardiomyocytes. Here, we first determine acoustic conditions suitable for the viable delivery of large proteins to hiPSCs with clinical Definity microbubble agents using our customized experimental platform. From here, we delivered Cas9-sgRNA RNP complexes targeting the EGFP (enhanced green fluorescent protein) gene to EGFP-expressing hiPSCs for EGFP knockout. Simultaneous acoustic cavitation detection during treatment confirmed a strong correlation between microbubble disruption and viable FUS-mediated protein delivery in hiPSCs. This study shows for the first time the potential for an FUS-mediated technique for targeted and precise CRISPR-Cas9 gene editing in human stem cells.

INTRODUCTION

CRISPR-Cas9 systems have quickly emerged at the forefront of gene editing and can be used to efficiently edit target gene sequences within the genome.^{1,2} As a result, CRISPR editing systems have been heavily considered for the treatment of genetic diseases (e.g., familial cardiomyopathy)^{3,4}; however, a lack of translatable delivery strategies has hindered their use for *in vivo* applications. Nevertheless, Cas9 is being employed across multiple ongoing clinical trials, mainly in an *ex vivo* context. Indeed, the first CRISPR-based medicine sold under the brand name CASGEVY was recently approved by the US Food and Drug Administration for the treatment of sickle cell disease and transfusion-dependent β -thalassemia.⁵

CRISPR systems can be used to precisely edit genes through different mechanisms (e.g., homology-directed repair), but for the purpose of simple gene loss of function (LOF), non-homologous end joining (NHEJ) is generally employed.⁶ CRISPR-associated protein 9 (Cas9) in complex with a single-guide RNA (sgRNA) works by first generating a double-stranded break (DSB) at a target site specified by the sgRNA. NHEJ occurs as a result of imperfect endogenous DNA repair mechanisms leading to the formation of short indels (insertions or deletions) at the targeted location, which can cause frameshift mutations in protein-coding regions. These frameshift mutations may lead to early gene termination (e.g., by generating an early stop codon), leading to an LOF of the targeted gene.⁷

Cas9 is perhaps most efficiently delivered to human cells with an sgRNA as a ribonucleoprotein (RNP) complex. This delivery format has been shown to provide high targeting efficiency,^{8,9} provide quick DNA cleavage post-transfection,¹⁰ and lack insertional mutagenesis relative to other strategies.¹¹ Current delivery methods for Cas9-sgRNA RNPs include microinjection, electroporation, liposomes, lipid nanoparticles, and viral delivery methods.^{12–14} Microinjection delivers Cas9 RNPs directly to the cell nucleus and is highly efficient; however, it provides low-throughput and is limited to *in vitro* transfection.^{15,16} Electroporation is commonly used for high-throughput gene editing and efficiently delivers Cas9 RNPs using pulsed high-voltage current across cell surfaces, but like other physical transfection methods, is generally limited for *in vitro* and *ex vivo* applications due to low tissue penetration depth and high toxicity.^{14,17} The use of liposomes and lipid nanoparticles is being explored for *in vivo* applications, but this poses the risk of unwanted immunogenic response and suffers from high levels of endosomal entrapment, hindering the efficiency of gene editing.^{18,19} Recently developed viral delivery methods such as the use of lentiviral capsid particles^{12,13} show promise but are viral in nature and can lead to unwanted immunogenic response. Adeno-associated viruses (AAVs) show perhaps the highest

Received 28 October 2024; accepted 7 January 2025;
<https://doi.org/10.1016/j.ymthe.2025.01.013>

Correspondence: Dr. Brandon Helfield, Department of Physics, Concordia University, 7141 Sherbrooke St. W, H4B 1R6 Montreal, QC, Canada.

E-mail: brandon.helfield@concordia.ca

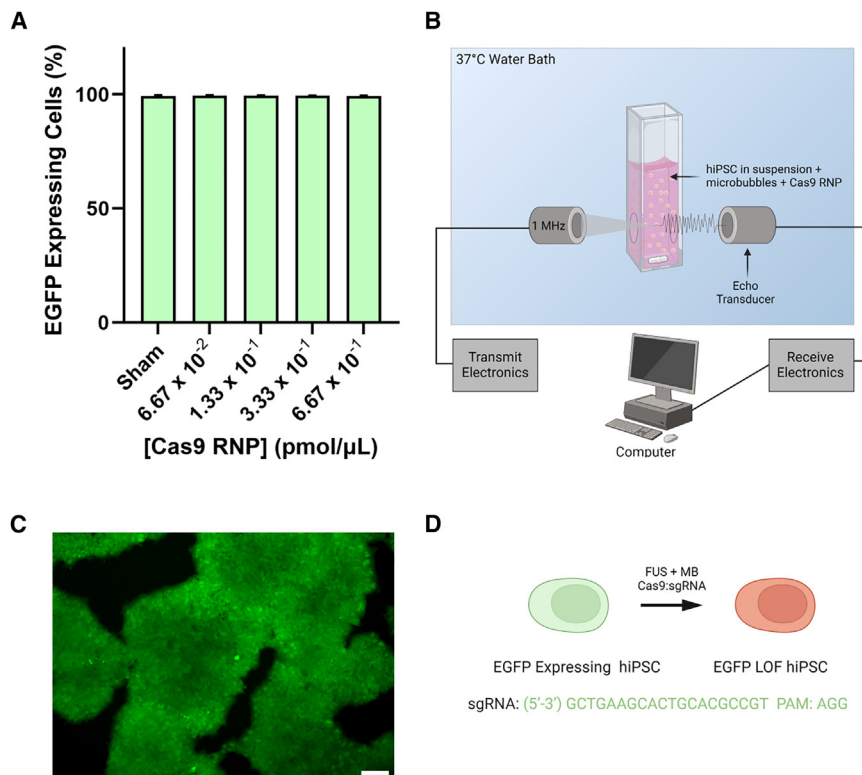


Figure 1. Cas9 RNPs cannot readily enter cells

(A) Varying concentrations of Cas9 RNP were incubated in suspension with 317-12-Ff hiPSCs for 10 min, including a sham control (incubated without Cas9-sgRNA RNP). Cells were recovered, and EGFP loss of fluorescence in the cell population was quantified on day 8 using flow cytometry to determine cell permeability to RNPs. One-way ANOVA followed by Tukey's post hoc test was used to determine that there was no significance between cell permeability and increasing concentrations of RNP (error bar, SEM). (B) Custom FUS setup used for 150-kDa FITC-dextran deliveries and Cas9 transfection. (C) EGFP-expressing 317-12-Ff hiPSCs (scale bar, 100 μm). (D) sgRNA targeted to EGFP LOF. No statistical difference between groups was observed unless noted.

potential for clinical applications of Cas9 gene therapy; however, they are limited by a small cargo capacity, making it difficult to deliver Cas9 RNPs due to their relatively large size (~160 kDa) and are therefore generally employed to deliver Cas9 as DNA plasmids.^{3,4} The lack of *in vivo* viable transfection methods provides an opportunity to develop a non-immunogenic modality for the delivery of Cas9 RNPs to treat genetic diseases.

Microbubbles are currently approved as contrast agents for use in ultrasound (US)-based imaging modalities (e.g., echocardiography), and under specific acoustic parameters can be utilized to transiently permeate cellular membranes.²⁰ Focused US (FUS) in combination with microbubbles may offer a viable alternative for *in vivo* delivery strategies of Cas9 RNPs for gene therapy due to its non-toxic image-guided nature. Microbubbles are small gas-filled lipid-encapsulated bubbles that, upon passing through the US beam, oscillate about their resting size and enhance the US echo from the vasculature to which they are confined. Indeed, localized microbubble vibrations and the associated generation of membrane perforation can allow for nearby otherwise impermeable macromolecules such as proteins to pass through treated tissue,^{21–23} as well as trigger local calcium influx,^{24,25} modulate ATP,²⁶ and stimulate gene expression.^{27–29} FUS and microbubbles are currently being explored as a local targeted drug/gene delivery platform with clinical trials in neuro-oncology applications,³⁰ as well as successful preclinical development within the context of cardiovascular disease.²² Simultaneously, there has been extensive research into genetic targets for cardiovascular disease by

employing human induced pluripotent stem cell (hiPSC)-derived cardiomyocytes as model systems.^{4,31,32}

Here, using clinical Definity contrast agent microbubbles, we show for the first time that FUS can be used to viably deliver large Cas9:sgRNA RNPs to hiPSCs *in vitro* as proof of concept, which can extend to downstream cardiovascular application. We optimize the FUS parameters, microbubble dose, and protein concentration conditions required to viably deliver the otherwise impermeable protein complex. We then demonstrate successful FUS-mediated delivery of CRISPR-Cas9 for gene editing and confirm microbubble echo characteristics responsible for such delivery as a clinically relevant readout for future translation.

RESULTS

Delivery of 150-kDa FITC-dextran into hiPSCs in suspension

Cas9 RNPs are highly efficient and have demonstrated low frequencies of off-target editing within the cells; however, their large size (~160 kDa) and negatively charged surface make them relatively impermeable to cells. To confirm their inability to readily enter cells, we incubated hiPSCs with increasing concentrations of EGFP-targeting Cas9-sgRNA RNPs and measured the frequency of EGFP loss 8 days post-incubation. We observed no significant difference in the percentage of EGFP-expressing cells across Cas9 RNP concentrations (Figure 1A). For physical transfection methods such as FUS-mediated delivery, the large cargo size is perhaps the greatest hurdle to overcome as it requires the US-induced surface pores to be at least as large as the diameter of the Cas9 protein. Despite evidence of small-cargo delivery (e.g., plasmid, small interfering RNA, microRNA), a limited literature exists on FUS and microbubble-mediated delivery of large macromolecules such as Cas9 RNPs. Therefore, as a surrogate for Cas9 RNP transfection, we first delivered a 150-kDa fluorescein isothiocyanate (FITC)-dextran protein to optimize US parameters, both to increase throughput and determine whether hiPSCs could

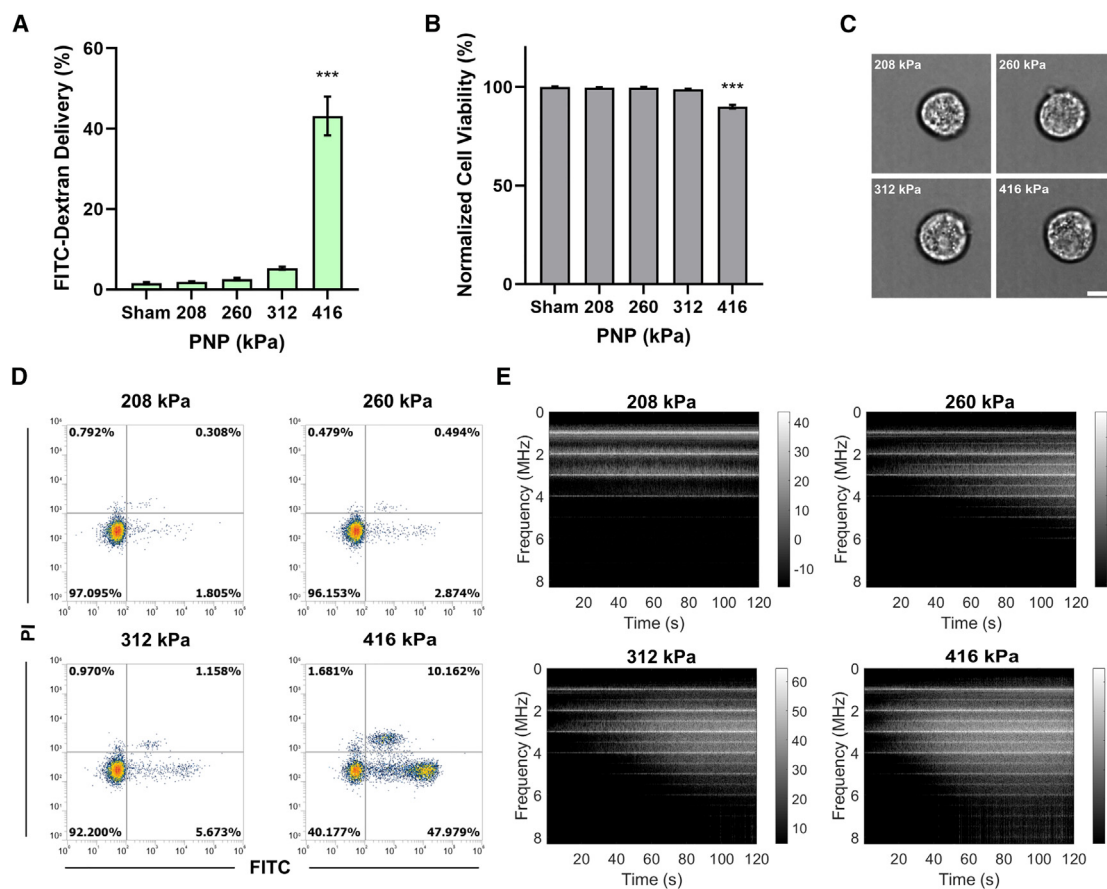


Figure 2. FUS and microbubbles can deliver 150 kDa FITC-dextran to hiPSCs

We co-incubated 5 mg/mL 150-kDa FITC-dextran with 1.5×10^6 201B7 hiPSCs and 3.75×10^8 Definity microbubbles and treated for 2 min under a US beam. Sham was incubated in the same amount of FITC-dextran and microbubbles but was left untreated for 2 min. (A) Higher PNPs show higher FITC-dextran delivery quantified by flow cytometry. (B) PI cell viability assay was used to quantify cell viability normalized to the sham control, where the highest pressure showed a significant decrease in viability relative to sham control and lower PNPs. (C) Representative single-cell images of hiPSCs using an Attune CytPix onboard camera. Imaging was used to gate around the hiPSC population for quality control (scale bar, 10 μ m). Cell size and morphology remained consistent throughout all pressures. (D) FITC/PI quadrant flow plots where FITC⁺/PI⁻ were used to quantify FITC delivery and PI⁺ was used to quantify cell viability. (E) Joint time frequency plots demonstrating microbubble disruption over the treatment time. An increasing amount of broadband signal over time (density of white dots) as a function of PNP can be observed. At lower pressures, no such broadband emission is observed at higher frequencies (4–8 MHz), with limited signal at lower frequencies (1–4 MHz) relative to higher pressures. At the highest pressure, 416 kPa, an increase in broadband emission is observed within 50 s into the treatment. For the statistical analysis, a one-way ANOVA followed by Tukey's post hoc test was used to determine the significance between samples (** $p < 0.001$).

survive long enough periods in suspension to be transfected efficiently.

As a starting point, we used parameters optimized previously by our lab for microRNA delivery to HUVE cells.³³ Samples consisting of 1.5×10^6 201B7 hiPSCs were co-incubated with Definity microbubbles at a 250-microbubble:cell ratio in a 5-mg/mL solution of FITC-dextran and treated in suspension for 2 min in our custom US setup (Figure 1B). We varied peak negative pressure (PNP), a critical parameter that strongly influences bubble vibration physics,³⁴ to find the optimal parameters for delivery. We found that higher PNP increased FITC-dextran delivery, while also increasing cell death (Figures 2A and 2B). At the highest pressure tested (416 kPa), FUS

and microbubbles yielded FITC-dextran delivery efficiencies of up to $41.6\% \pm 4.84\%$, while also leading to the lowest cell viability ($90.1\% \pm 0.9\%$ normalized to the sham control, $p < 0.001$; Figures 2A–2D). Interestingly, at lower pressures, differences in FITC-dextran delivery were non-significant compared to the sham control, with negligible differences in cell viability. During the treatment of each sample, it was observed that microbubbles exposed to the highest PNP would undergo bubble collapse completely by the 2-min treatment time. In contrast, microbubbles at lower pressures did not undergo such obvious disruption. This acoustic behavior was quantified via passive cavitation detection and shown here by the use of joint time-frequency panels (Figure 2E). These show the acoustic spectrum (the frequency of the microbubble

scattered pressure) plotted against treatment time, with the grayscale denoting the magnitude of the bubble scattering. It can be seen here that along with a strong fundamental component (the treatment frequency, $f = 1$ MHz), there exists stable scattering at harmonic components (nf , where $n = 2, 3, 4 \dots$) throughout the treatment time. With increasing pressure, broadband emissions (scattering occurring between the harmonics) become apparent as treatment progresses, occurring at approximately 60 s for 312 kPa and as early as 20 s for the strongest treatment condition (416 kPa; Figure 2E). Indeed, the presence of such broadband emissions are a well-known indicator of microbubble disruption, known as inertial cavitation.³⁵ It was therefore hypothesized that microbubble disruption may have had a significant effect on FITC-dextran delivery and perhaps led to larger pore formation, allowing for the delivery of the protein.

To explore this hypothesis, we decreased the microbubble dose while keeping the other FUS and sample parameters fixed to determine whether multiple-bubble scattering and subsequent US attenuation may play a role in inertial cavitation and subsequent protein delivery.³⁶ The microbubble dose was halved to 125 microbubbles per cell, and samples were treated with the same PNP along with an additional sample set at 364 kPa to fill in an observable gap in the acoustic pressures used. The decrease in microbubble dose increased delivery across all PNPs significantly relative to sham while at the same time maintaining similar cell viabilities (Figures 3A and 3B). FITC-dextran delivery was quantified by flow cytometry and ranged from $9.309\% \pm 0.52\%$ at the lowest pressure of 208 kPa up to $50.46\% \pm 1.17\%$ at 416 kPa (Figure 3A). When normalized to the sham to control to account for FITC-dextran binding to the exterior surface of the cell or passive entry across the membrane, delivery ranged from $6.24\% \pm 0.52\%$ to $47.38\% \pm 1.17\%$ at 208 and 416 kPa, respectively (Figure 3A). As the main objective of the surrogate experiment was to optimize protein delivery while maintaining high cell viability, the PNP of 364 kPa was chosen for subsequent surrogate deliveries to further optimize parameters for FUS and microbubble-induced Cas9 RNP transfection. Between the two highest PNPs, the non-significant change in sham-normalized FITC-dextran delivery with a significant decrease in cell viability (Figure 3B) led to this decision. The treatment at 364 kPa yielded a high sham-normalized delivery of $42.24\% \pm 1.89\%$ while maintaining a relatively high cell viability of $94.54\% \pm 0.80\%$.

FITC-dextran titration assay and multiple FUS treatments

With the microbubble dose fixed at 125 microbubbles per cell, an FITC-dextran titration assay determined that a concentration of 6.7×10^{-1} pmol/ μ L was the lowest concentration with no significant difference in FITC-dextran delivery to hiPSCs. FITC delivery at 6.7×10^{-1} pmol/ μ L rivaled the observed delivery of higher FITC-dextran concentrations at $31.7\% \pm 0.34\%$ relative to $37.33\% \pm 3.35\%$ and $38.03\% \pm 0.38\%$ at 6.7 and 33 pmol/ μ L, respectively (Figure 4A).

We then repeated our initial experiments to determine differences in FITC-dextran delivery at varying pressures with this optimized lower

concentration of FITC. At 364 kPa with a 6.7×10^{-1} pmol/ μ L solution of FITC-dextran, delivery efficiency only decreased to $37.90\% \pm 0.31\%$ (Figure 4B), with a 50-fold decrease in FITC-dextran concentration (see Figure 3). Similar decreases were observed across all PNPs while maintaining the strong linear relationship between treatment pressure and observed FITC-dextran delivery. Additionally, we were interested in whether repeating treatments multiple times at a lower PNP with near-100% cell viability may yield more efficient delivery than a single treatment at higher PNPs. Such multiple treatments at the lowest pressure of 208 kPa shows interesting results, with significant increases from $5.39\% \pm 2.9\%$ delivery after one treatment to $8.20\% \pm 0.25\%$ ($p < 0.01$) while maintaining cell viability (not significant; Figures 4C and 4D). These results provided strong evidence that FUS in combination with microbubbles could be utilized to deliver a large protein (FITC-dextran, 150 kDa) efficiently to hiPSCs in suspension, with delivery efficiencies as high as $37.90\% \pm 0.31\%$ while maintaining consistently high cell viabilities ($\sim 95\%$ at a PNP of 364 kPa).

FUS-induced Cas9 RNP transfection

Following the surrogate experiments, we next delivered Cas9:sgRNA RNP complexes (1:3 M ratio) at 6.7×10^{-1} pmol/ μ L at a 1-MHz frequency, 1,000 burst cycles, 5-ms pulse intervals to EGFP expressing 317-12-Ff hiPSCs (Figure 1C) in suspension with microbubbles at 125 microbubbles per cell (1.5×10^6 cells). We used Cas9-sgRNA RNPs targeting EGFP (Figure 1D) to induce indels, using loss of EGFP fluorescence as a readout. Editing efficiency was measured 8 days post-transfection with flow cytometry using non-EGFP-expressing hiPSC 201B7 as a positive control for the EGFP LOF. EGFP LOF was measured at $10.57\% \pm 0.16\%$ when normalized to sham (here, non-treated EGFP-expressing hiPSCs) to control for non-EGFP-expressing cells and random loss of fluorescence within the population (Figures 5A–5D). A cell viability assay was performed immediately after Cas9 transfection with an observable and significant decrease in cell viability in the treated sample at $77.18\% \pm 0.60\%$ relative to sham (Figure 5D). To account for the change in cell viability post-transfection, we also performed a 72-h 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) viability assay, which demonstrated a drop in cell viability in treated cells to $64.87\% \pm 0.82\%$ (Figure 5C). These decreases in cell viability are notable; however, they may be due to the cytotoxic effects of Cas9 DSBs, an issue not previously seen with FITC-dextran delivery due to its inert nature.

DISCUSSION

In this study, we demonstrated for the first time that FUS and microbubbles can be used to deliver Cas9 RNPs to hiPSCs, undifferentiated human stem cells with the ability to differentiate into a variety of cell types (e.g., cardiomyocytes), as proof of concept for FUS-mediated CRISPR gene editing therapy with RNPs. We show FUS-mediated delivery of large protein surrogates (150 kDa) similar in size to Cas9 RNPs into hiPSCs, with upward of $42.24\% \pm 1.89\%$ delivery while maintaining high cell viabilities of $94.54\% \pm 0.80\%$

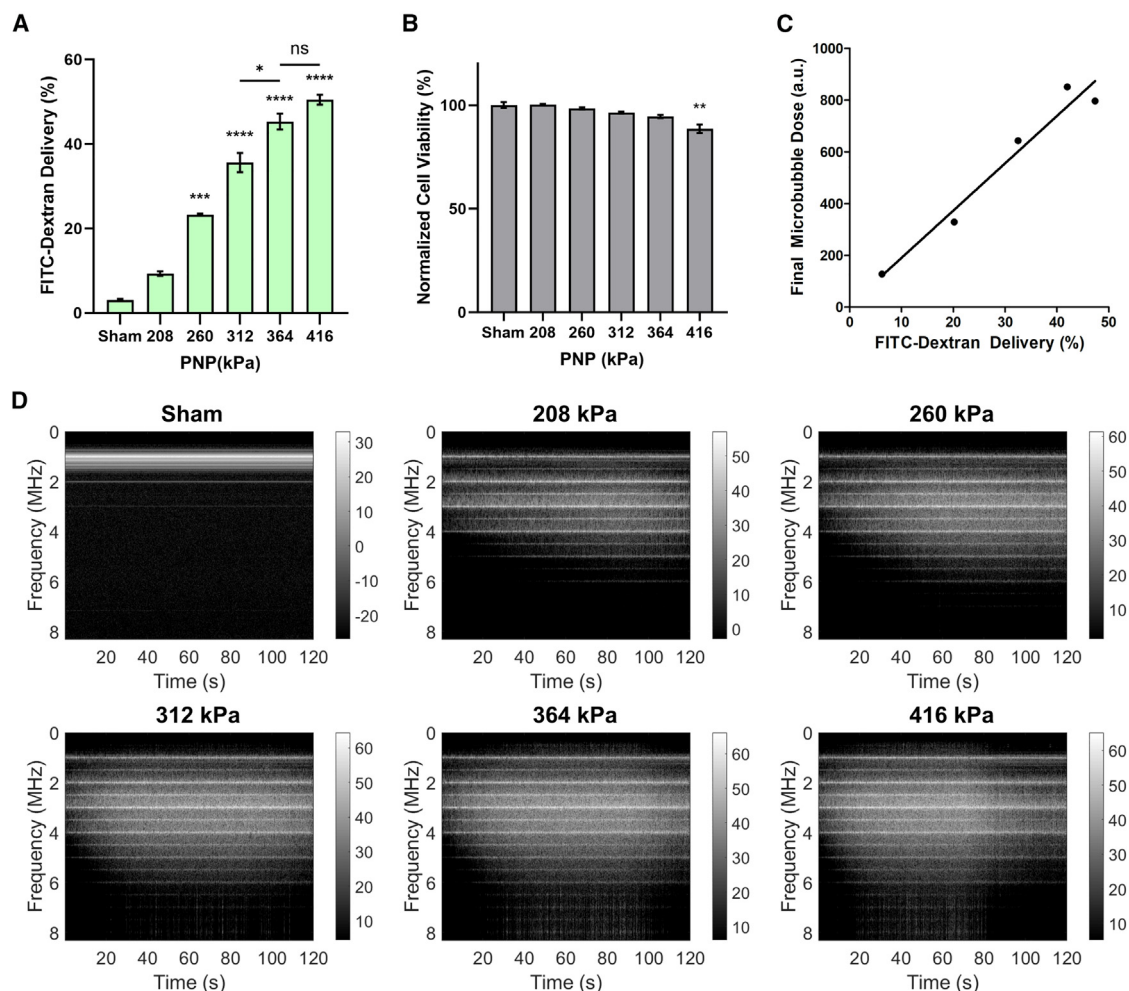


Figure 3. Lowering microbubble dose increases FITC-dextran delivery

We co-incubated 5 mg/mL of 150 kDa FITC-dextran with 1.5×10^6 201B7 hiPSCs and 1.875×10^8 Definity microbubbles and treated them for 2 min under a US beam. Sham was incubated in the same amount of FITC-dextran and microbubbles but left untreated for 2 min. (A) Higher PNPs show increased FITC-dextran delivery with a significant increase relative to sham control starting at 260 kPa. No significant change in FITC expression can be observed between 364 and 416 kPa. (B) PI viability assay immediately after US treatment shows non-significant changes in cell viability relative to sham up to 364 kPa. A significant decrease in cell viability is observed at a PNP of 416. (C) Cumulative inertial cavitation dose (final microbubble dose) as a function of FITC⁺ viable cell percentage. Linear regression was used to measure the correlation between microbubble disruption (broadband emissions) and FITC expression in US-treated cells ($R^2 = 0.956$, $p < 0.001$). (D) Scattered pressure emissions from microbubbles over time under US treatment at different pressures. A one-way ANOVA followed by Tukey's ad hoc test was used to determine significance in (A) and (B) (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$). Error bars, SEMs.

(not significant; Figures 3A and 3B), efficiency levels of which correlate with FUS-induced bubble disruption.

To achieve these high delivery efficiencies and cell viabilities, we focused our efforts on modifying PNP and microbubble dose, two key parameters that can have a large effect on both delivery efficiency and cell viability. Specifically, the mechanical index (defined as the PNP divided by the square root of transmit frequency) is well established to have a significant impact on microbubble vibration³⁷ and related bioeffects.³⁸ Here, with our transmit frequency fixed to 1 MHz, this translates to the PNP, whereby microbubble oscillations

become larger and increasingly lead to bubble collapse with increasing PNP, all else being equal. Although the mechanism of delivery has not yet been fully elucidated, higher PNPs likely lead to larger cell membrane perforations, perhaps leading to increased cell death due to inefficient membrane re-sealing.²¹ High PNPs (e.g., 416 kPa) delivered higher levels of 150-kDa FITC-dextran than lower PNPs (Figure 2A), but compromised cell viability in the process (Figure 2B). For this reason, we opted to choose a slightly lower PNP of 364 kPa, which had a non-significant change in delivery efficiency, but a significant increase in cell viability ($p < 0.01$; Figures 3A and 3B). Microbubble dose also played a key role in increasing delivery

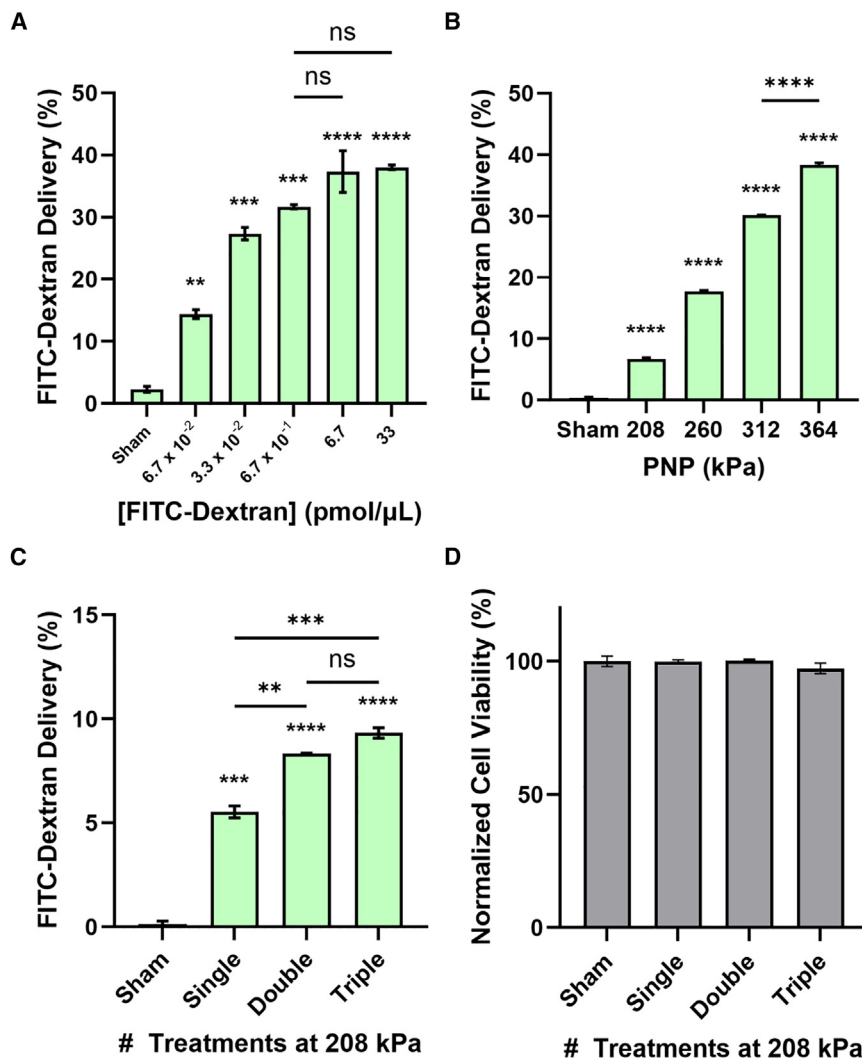


Figure 4. Surrogate large protein can be delivered at lower concentrations over multiple treatments while preserving cell viability

We titrated 150 kDa FITC-dextran to determine whether lower concentrations (6.7×10^{-2} pmol/μL) yielded similar levels of FITC-dextran delivery relative to higher concentrations (e.g., 33 pmol/μL). (A) A concentration range was delivered at 364 PNP, using flow cytometry to quantify FITC expression. Higher concentration yielded increased FITC expression relative to sham control with no significant difference observed past 6.7×10^{-1} ($p > 0.05$). (B) FITC-dextran at 6.7×10^{-1} pmol/μL was delivered at increasing PNPs to determine the pressure at which the largest delivery is observed without compromising cell viability. (C and D) A PNP of 208 kPa was used for multiple treatments to determine whether re-treatment of hiPSCs shortly after the initial treatment may yield higher FITC delivery while maintaining higher cell viability than under a single treatment at a higher PNP. Multiple treatments showed a significant increase in FITC-dextran delivery relative to a single treatment while yielding no significant difference in cell viability based on PI assay immediately after treatment. An ANOVA followed by Tukey's post hoc test was used to determine the significance between samples and sham and between each other (** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$). Error bars, SEMs.

efficiencies. By reducing microbubble dose from 250 to 125 MB/cell, we in turn increased delivery efficiencies across all PNPs used (Figures 2A and 3A). Based on both observational evidence of a lack of microbubble disruption and the lack of broadband emissions from lower PNPs demonstrated as time-frequency plots (Figure 2E), we determined the high microbubble dose at 250 MB/cell was likely attenuating the FUS signal. This was especially apparent at lower pressures, which demonstrated little broadband echo during treatment (Figure 2E). This in turn reduced bubble disruption, which we found to be important as demonstrated by the positive correlation between delivery and broadband emissions (Figures 3C and 3D).

Using parameters determined at the surrogate level, we next delivered Cas9-sgRNA RNPs targeting the EGFP gene to EGFP-expressing 317-12-Ff hiPSCs and achieved editing efficiencies of >10% (Figures 5A, 5D, and 5E). It is important here to place this efficiency within the greater landscape of Cas9-mediated editing via NHEJ. Cas9-mediated modification of transformed cells through NHEJ typically reaches

rates of ~20%–90%^{8,39–43} and is markedly less effective in the particular context of human PSCs (from 5- to 20-fold less),^{44,45} which places our Cas9-mediated NHEJ-based editing on par with the current framework. Interestingly, the high cell viability observed at the surrogate stage did not translate to the Cas9 transfections. Although Cas9 RNPs are highly efficient, transient, and less cytotoxic relative to other Cas9 systems, they can have an immediate cytotoxic

effect upon entry to the nucleus of the cell due to the induced DSBs required for NHEJ, in part led by a p53-activated DNA damage response.^{46,47} The notable decrease in cell editing efficiency relative to delivery of surrogate protein could also therefore be explained by the increased cell death observed up until 72 h post-transfection. Indeed, while there is only a limited literature on the use of FUS for CRISPR gene editing, other groups have shown similar efficiencies in different model systems using alternative CRISPR machinery. Very recently, Lao and colleagues used FUS to open the blood-brain barrier for the administration of CRISPR-encoding AAVs, where they achieved editing efficiencies >25% in specific brain regions in C57BL/6 mice.⁴⁸ Additionally, Anderson et al. have demonstrated that FUS could be used to deliver cytosine CRISPR-based base editing plasmid constructs targeting the *Pde3b* gene in adult mouse livers, where they achieved ~2.5% on-target editing.⁴⁹ Despite the exciting progress demonstrated in FUS-mediated delivery of CRISPR machinery, the use of CRISPR-based base editors⁵⁰ presents a promising next step for FUS-induced gene editing as they can introduce mutations in a

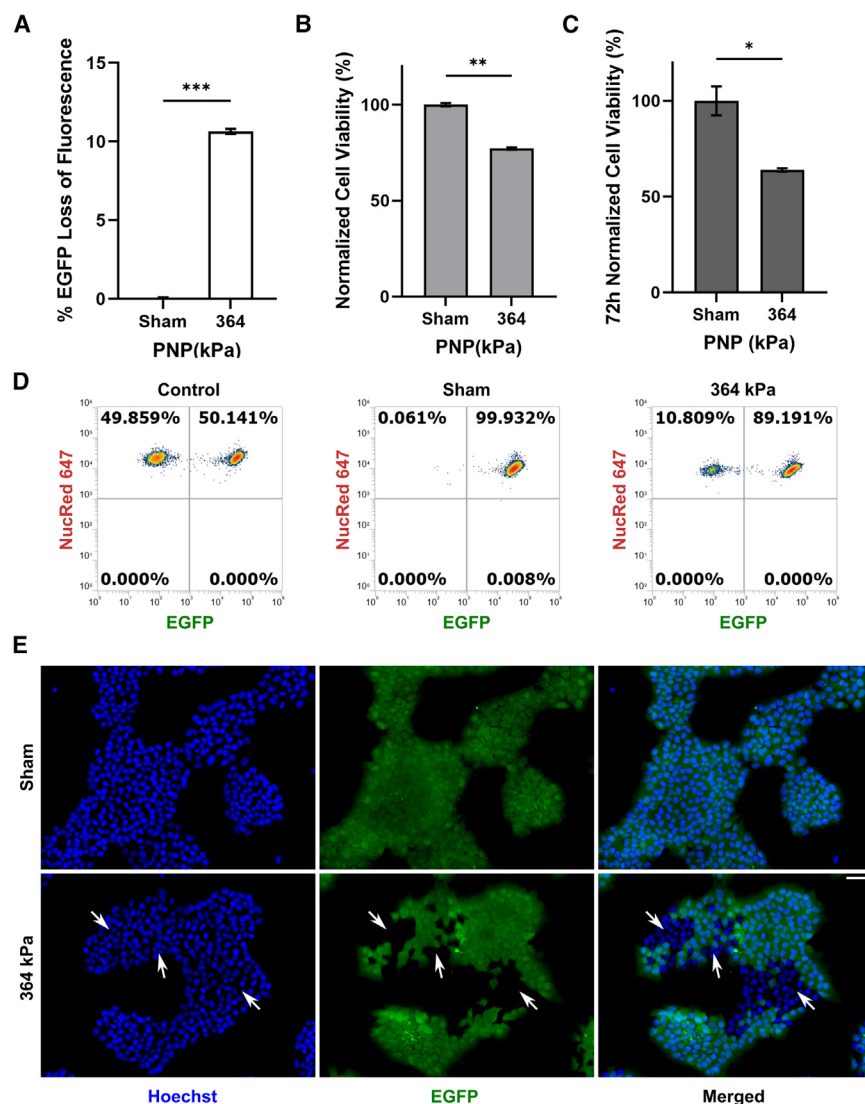


Figure 5. FUS and microbubbles can transfect Cas9 RNPs for EGFP LOF

EGFP-expressing 317-12-Ff hiPSCs were co-injected with Definity microbubbles and 6.7×10^{-1} pmol/ μ L Cas9:sgRNA RNP at a 1:3 M ratio (sham control = microbubble only, no US) and treated with US at 364 kPa with fixed parameters. (A) Sham-normalized EGFP LOF assay. Approximately $\sim 1.0 \times 10^5$ cells from each sample were reseeded on 6-well plates and allowed to recover for 8 days before EGFP expression quantification. Treated sample EGFP expression was quantified using flow cytometry on day 8 relative to sham control to determine EGFP LOF. A 1:1 cell ratio of non-EGFP-expressing 201B7 hiPSCs and non-treated 317-12-Ff were used to gate for EGFP with NucRed 647 as a nuclear stain. (B) Sham-normalized PI viability assay. The assay was performed immediately after transfection to determine cell viability associated with treatment. Significant loss of viability in the 364-kPa treated sample was observed relative to sham control. (C) Sham-normalized 72-h MTT cell viability assay. A subset of hiPSCs from the initial treatment was kept to determine post-treatment cell viability. Cells were concentration matched, and absorbance measurements were taken 72 h after treatment. A significant decrease in cell viability is observed in the treated sample relative to sham control. (D) Representative flow cytometry plots demonstrating a significant shift in EGFP expression (top left quadrant, NucRed⁺/EGFP⁺). Control was a 50%:50% cell suspension of 201B7:317-12-Ff cells. (E) EGFP loss of fluorescence can be observed with fluorescence microscopy. All cells were imaged at 200 $\times$ total zoom under the same settings (scale bar, 50 μ m). White arrow indicates patches of cells with loss in EGFP expression. A two-way Student's t test was used to determine significance (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). Error bars, SEMs.

precise window without inducing a DSB. Base editing involves the use of catalytically impaired Cas9 proteins with inactivated endonuclease domains (e.g., nCas9) fused to deaminase proteins (e.g., adenine deaminases, cytidine deaminase). The catalytically impaired nCas9 in complex with sgRNA acts as a targeting mechanism, which allows the fused deaminase enzymes to catalyze 4 of the 12 possible base substitutions at the site of interest (C-to-T, T-to-C, G-to-A, and A-to-G).⁵¹ The lack of endonuclease activity and fused base editor circumvents the requirement for a DSB to induce a genomic edit along with the associated cytotoxicity. Newer and more efficient precision editing tools such as CRISPR-based prime editors utilize an inactivated Cas9 protein fused to a reverse transcriptase (RT) enzyme, and an extended gRNA called prime editing guide RNA (pegRNA) to cause genomic edits without the requirement of a DSB. The fused RT allows for the extended RNA domain of the pegRNA RT to be copied into the targeted DNA sequence. Prime editors can in turn

mediate all 12 base-base conversions as well as targeted indels in a genomic sequence.⁵² However, their large size has limited delivery as an RNP—perhaps a limitation that could be overcome with FUS and microbubbles. Nevertheless, the high efficiency and low cytotoxicity of prime editors offers a highly efficient alternative and already has been established is a variety of hard-to-transfect human cell lines such as primary human T cells⁵³ and hiPSCs.⁵⁴

To this point, Cas9 gene therapy has been heavily considered for the treatment of cardiac genetic disorders such as hypertrophic cardiomyopathy and dilated cardiomyopathy. Due to a limited number of alternative treatment strategies, its use remains pre-clinical because of concerns surrounding potential *in vivo* viable delivery methods. Recent attempts to deliver Cas9 base editors to human cardiomyocytes⁴ and mice³ for the treatment of hypertrophic cardiomyopathy, a common cardiac genetic disease generally caused by single

nucleotide polymorphisms (SNPs), have been successful, but rely on the use of AAV9 for delivery. Additional attempts by other groups using Cas9 base editing have proven successful in correcting dilated cardiomyopathy, another cardiac disease generally associated with pathogenic SNPs.⁵⁵ With newer CRISPR technologies in the developmental pipeline and the inherent need for a non-invasive *in vivo* viable delivery method for the treatment of diseases that cannot be treated extra-corporeally, parameters developed here in an *in vitro* context may translate to a viable alternative for *in vivo* applications to current technologies. As the end goal would be to apply this technology in an *in vivo* context, we believe editing efficiencies of >10% in an hiPSC line *in vitro* were sufficient as proof of principle for future investigations. However, toxicity concerns relating to Cas9 itself will need to be addressed before optimal FUS-mediated delivery can occur *in vivo*. With certain clinical US systems having simultaneous imaging and treatment capabilities, this would allow for real-time feedback of Cas9 delivery to heart tissue. This feature of US-guided therapy would have major advantages over current *in vivo* viable Cas9 delivery strategies (e.g., AAV) as it would allow for physical targeting of diseased tissue. Perhaps most exciting are the results obtained over multiple treatments of hiPSCs with lower pressures (Figures 4C and 4D). Here, we show that FUS can be used within a repeated treatment paradigm whereby lower PNPs can be utilized to effectively deliver large proteins in smaller quantities without compromising cell viability. Indeed, this repeat-treatment technique would be made possible by the non-invasive and image-guided nature of FUS.

This study aimed to demonstrate that FUS in combination with microbubbles could help to solve the issue of Cas9 RNP delivery for genetic disease correction, by developing an FUS system capable of transfecting Cas9 RNPs to hiPSCs, a cell type with huge potential in genetic cardiac disease modeling. FUS and microbubbles may offer a viable Cas9 RNP delivery strategy with clinical applications in gene therapy. FUS is a non-toxic modality that is inherently image guided, and the use of Definity microbubbles is already used clinically for contrast imaging (e.g., echocardiography). The non-invasive nature of FUS would allow for targeted delivery of Cas9 RNPs to a specific tissue such as left ventricular cardiomyocytes for disease correction.

MATERIALS AND METHODS

Cell culture

The 201B7 hiPSCs⁵⁶ were obtained from American Type Culture Collection. The 317-12-Ff hiPSCs (Figure 1C) were generously provided by Dr. Knut Woltjen from the Center of iPS Cell Research and Application (CiRA).⁵⁷ hiPSCs were cultured in 6-well plates matrix coated in either Corning Matrigel or iMatrix-511 Silk (Nippi) as per the manufacturer's protocol in mTeSR Plus or eTeSR media (STEMCELL Technologies). Cells were kept at 37°C, 5% CO₂, and passaged every 3–4 days at 80%–90% confluency by dissociating them with Accutase (STEMCELL Technologies) according to the manufacturer's protocol. After dissociation, cells were resuspended and ~100,000 cells were transferred into freshly coated plates, with 2 mL mTeSR Plus or eTeSR media containing 10 μ M Y-27632

ROCK inhibitor (STEMCELL Technologies). After 24 h, old media was replaced with 2 mL fresh media and replaced every 1–2 days until the following passage.

Ultrasound apparatus and setup

A custom-focused US treatment tank was employed for all transfections. The treatment tank consisted of a 30-L plastic container with two co-aligned single-element transducers and an acrylic treatment cuvette centered between the transducers. Before experiments, the tank was filled with ~15 L gas-equilibrated distilled H₂O, enough to cover both transducers, but kept below the opening of the treatment cuvette to reduce the risk of sample contamination. The water temperature was maintained at 37°C using an immersion heat circulator (VWR model 1120, VWR International). A magnetic stir bar was placed inside the treatment cuvette sealed with Mylar windows (25 μ m thickness) to allow ultrasonic transmissions. The treatment cuvette was placed above a magnetic stir plate to allow samples to mix throughout treatment (Figure 1B).

FUS and microbubble-mediated FITC-dextran delivery

The 201B7 cells were resuspended in Accutase (STEMCELL Technologies) per the manufacturer's protocol. Cells were centrifuged at 500 rpm for 5 min at room temperature and resuspended in 5 mL eTeSR media for counting. We added 20 μ L cell suspension to 20 μ L trypan blue (Gibco) for dead cell exclusion and counted with Invitrogen Countess 3 Automated Cell Counter. We suspended 1.5×10^6 cells in a total volume of 3 mL eTeSR media per sample and kept at room temperature until treatment. Microbubble vials were prepared by incubating at room temperature for 15 min before shaking in a vial shaker for 45 s, per the manufacturer's instructions. Microbubbles were allowed to cool at room temperature for 15 min after activation before being inverted for 2 min and extracted using a 19G PrecisionGlide syringe needle (no. 305196, BD Biosciences).⁵⁸ For all experiments, Definity microbubbles were used within 1 day of activation. A 25-mg/mL 150-kDa FITC-dextran solution in eTeSR was used as stock and added to cells with microbubbles in the treatment cuvette. The cell mixture was allowed to stir for 30 s prior to treatment, allowing for a homogeneous distribution of microbubbles and cells. Samples were then treated with FUS using treatment transducer ($f = 25.4$ mm, $f\# = 1.33$) at 1 MHz with 1,000 cycles and 5-ms pulse repetition intervals (duty cycle of 20%). PNPs ranging from 208 to 416 kPa were generated from an arbitrary function generator (AFG31000, Tektronix) and amplified using an RF power amplifier (model no. 105A100B, Amplifier Research). Acoustic pressure was calibrated in a separate water tank using a hydrophone (ONDA). Post-treatment, cells were allowed to recover at room temperature for 5 min before being put on ice until washing.

Cas9:sgRNA RNP generation and treatment

SpCas9-2NLS and synthetic sgRNAs were purchased from Synthego. A Cas9:sgRNA RNP complex was assembled per the manufacturer's protocol. Briefly, for each FUS-treated sample, 6.0 nmol of lyophilized synthetic sgRNA was spun down and resuspended in 60 μ L $1 \times$ TE (Tris-EDTA) buffer, pH 8.0. We added 140 μ L nuclease-free water

to an sgRNA solution and incubated at room temperature for 5 min. After 5 min, 2.0 nmol Cas9 was added to the sgRNA solution and allowed to incubate for a minimum of 10 min prior to transfection. Samples were prepared and treated as previously described above. Post-treatment, cells were centrifuged at 500 rpm for 5 min at room temperature and washed two times in eTeSR before counting. Cells were recounted using Invitrogen Countess 3 Automated Cell Counter as described above. We plated 1.0×10^5 cells of each sample in iMatrix-511-coated 6-well plates for day 8 flow cytometry and microscopy. Remaining cells were used for propidium iodide (PI) cell death assay and 72-h MTT cell viability assay.

PI cell viability assay

Immediately after FUS treatment, a subset of the treated cells was allowed to recover for 5 min before being put on ice until centrifugation. Cells were then centrifuged at 500 rpm for 5 min at 4°C and washed twice in eTeSR media and once in PBS (without calcium and magnesium; Wisent) before being resuspended in 1.5 mL PBS and passed through a 35- μ m cell strainer to remove large cell clumps. Cells were kept on ice until flow cytometry was performed. A total of 1.5 μ L PI (final concentration of 1 μ g/mL) was added to each sample, vortexed, and incubated for 1 min before being analyzed with flow cytometry. All flow readouts were performed using an Attune CytPix (Thermo Fisher Scientific). For FITC-dextran surrogate transfections to 201B7 cells, voltages to gate for FITC and PI were set using a negative control or sham control for each experiment. FITC⁺/PI[−] cells were gated and normalized to sham control. PI⁺ cells were used to determine cell death associated with each FUS condition. For FUS transfection of Cas9 to 317-12-Ff cells for EGFP LOF, voltage to gate for PI was set using sham control with and without PI. Flow cytometry plots and images were analyzed using Attune NxT software and edited in Inkscape and Fiji-ImageJ, respectively.

MTT cell viability assay

A 4-h MTT cell proliferation assay was used to measure cell viability 72 h post-Cas9 RNP transfection. A subset of cells post-transfection was washed twice in eTeSR and seeded at a density of 5,000 cells/well of an iMatrix-511-coated 96-well plate in triplicate and maintained using the above cell culture protocol until assay. At 72 h post-seeding, old media was removed and washed three times with PBS. A 100- μ L 1.2- μ M MTT eTeSR solution was added to each well and incubated at 37°C for 4 h. After 4 h, 25 μ L MTT supplemented media was removed from each well. We added 50 μ L DMSO to each well and allowed them to incubate for 10 min at 37°C. After incubation, absorbance at 540 nm was measured using a Varioskan LUX microplate reader (Thermo Fisher Scientific) and analyzed to determine cell viability.

EGFP LOF assay

To test for FUS-mediated transfection of hiPSCs, an EGFP LOF assay was utilized. We used 317-12-Ff cells expressing EGFP at the AAVS1 locus for the assay. Post-transfection with Cas9:sgRNA RNP, cells were allowed to recover 5 min at room temperature before being placed on ice until centrifugation. Cells were centrifuged at

500 rpm for 5 min at 4°C and washed twice in eTeSR. Approximately 1.0×10^5 cells were concentration matched and seeded on iMatrix-511 (Nippi)-coated 6-well plates in eTeSR media supplemented with 10 μ M Y-27632 ROCK inhibitor and allowed to recover for 4 days. After 4 days, the cells were resuspended using Accutase and passed on to fresh iMatrix-511-coated 6-well plates in duplicate for day 8 flow cytometry and microscopy. Day 8 cells were resuspended using Accutase and centrifuged at 500 rpm. Following centrifugation, cells were resuspended in 1.5 mL PBS for flow analysis. For flow cytometry, the Attune CytPix was used to measure the change in EGFP expression across samples. Voltages to gate for EGFP were set using a 1:1 201B7:317-12-Ff cell solution. EGFP cells were gated and normalized to sham control. Flow cytometry plots were analyzed using Attune NxT software and edited in Inkscape.

Microscopy

All cell images were taken with an EVOS M7000 inverted fluorescence microscope (Invitrogen). Separate plates of iMatrix-511-coated 6-well plates were imaged 8 days post-transfection. Cells were washed once in PBS before being incubated with a 1:1,000 dilution of Hoechst nuclear stain in 1 mL eTeSR media. Before imaging, cells were washed three times in PBS and imaged in eTeSR media. During flow cytometry, single-cell images were captured with Attune CytPix imaging software to ensure there was no significant change in cell size or morphology.

Statistical analysis and Figures

All statistical analysis was performed using GraphPad Prism version 10.3.1. For FITC-dextran experiments, a one-way ANOVA followed by Tukey's post hoc test was used to determine significance between samples. All experiments were conducted in duplicate ($n = 2$). A two-way Student's *t* test was used to determine significance across samples for EGFP loss of fluorescence ($n = 2$). The graphs were generated using GraphPad Prism version 10.3.1.

DATA AND CODE AVAILABILITY

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

ACKNOWLEDGMENTS

The author(s) declare that financial support was received for the research, authorship, and/or publication of this article. This work was supported by the Canada Research Chairs program (CRC-2023-00137) and the Canadian Institutes of Health Research (CIHR-496216). B.H. holds a CASI award from the Burroughs Wellcome Fund (1018212.03). We thank Dr. Knut Woltjen from CiRA for generously providing the hiPSCs. We would also like to thank the Genome Foundry at Concordia University and Angela Quach for the help provided with hiPSC culture.

AUTHOR CONTRIBUTIONS

Conceptualization, B.H. Methodology, K.H., D.S., S.H., Z.G., M.C.H., and B.H. Validation, K.H. Investigation, K.H. Data curation, K.H. Formal analysis, K.H. and B.H. Visualization, K.H. Writing – original draft, K.H. and B.H. Writing – review & editing, K.H., M.C.H., and B.H. Supervision, B.H. Funding acquisition, B.H.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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